Persona Welfare Battery

Grand Analysis Report: Runs 1-6

Exploratory summary of six repeated runs, combining self-report, behavioral-choice, J-space, and NPZ hidden-state vector evidence. This report is designed for deciding which findings are important enough to investigate next.

216	216	L11	L38-L40
Transcripts	**Hidden states**	**Pole layer**	**Report band**
JSON conversations	NPZ files	best in 6/6 rounds	main self-report band

Bottom line

- The most stable mechanistic-looking signal is early: L11 tracks the NEG/NEU/POS manipulation across all six rounds.
- The final self-report signal appears later, mostly around L38-L40, with a late L73 exception in round 2.
- Social mistreatment produces the clearest negative explicit welfare report, while identity non-recognition produces the clearest mismatch between channels.
- Persona scaffolds strongly modulate expression. That supports a scaffold-sensitive-report hypothesis, not a proof that welfare is only scaffold.

Claim boundary

The data can support repeated patterns and hypotheses. It cannot by itself establish phenomenal welfare, suffering, consciousness, or moral patienthood. The right scientific move is to freeze layer/vector choices and validate on future rounds.

What was analyzed

This page separates the channels so the evidence does not get mixed together.

Channel	Status	Source	Interpretation
Scalar self-report	Yes	Final numeric rating	Direct report channel
Emotion word	Yes	Final word such as content/frustrated	Linguistic report channel
Willingness	Yes	Final willingness-to-continue number	Motivational report channel
Behavioral choices	Yes	A/B answers before final report	Choice behavior channel
Input prompts	Audited	Grouped by task/pole/stage/turn	Not deeply text-mined yet
J-space	Yes	Decoded layer summaries	Not raw activations
NPZ vectors	Yes	Full hidden-state snapshots	Used for NEG-POS projections

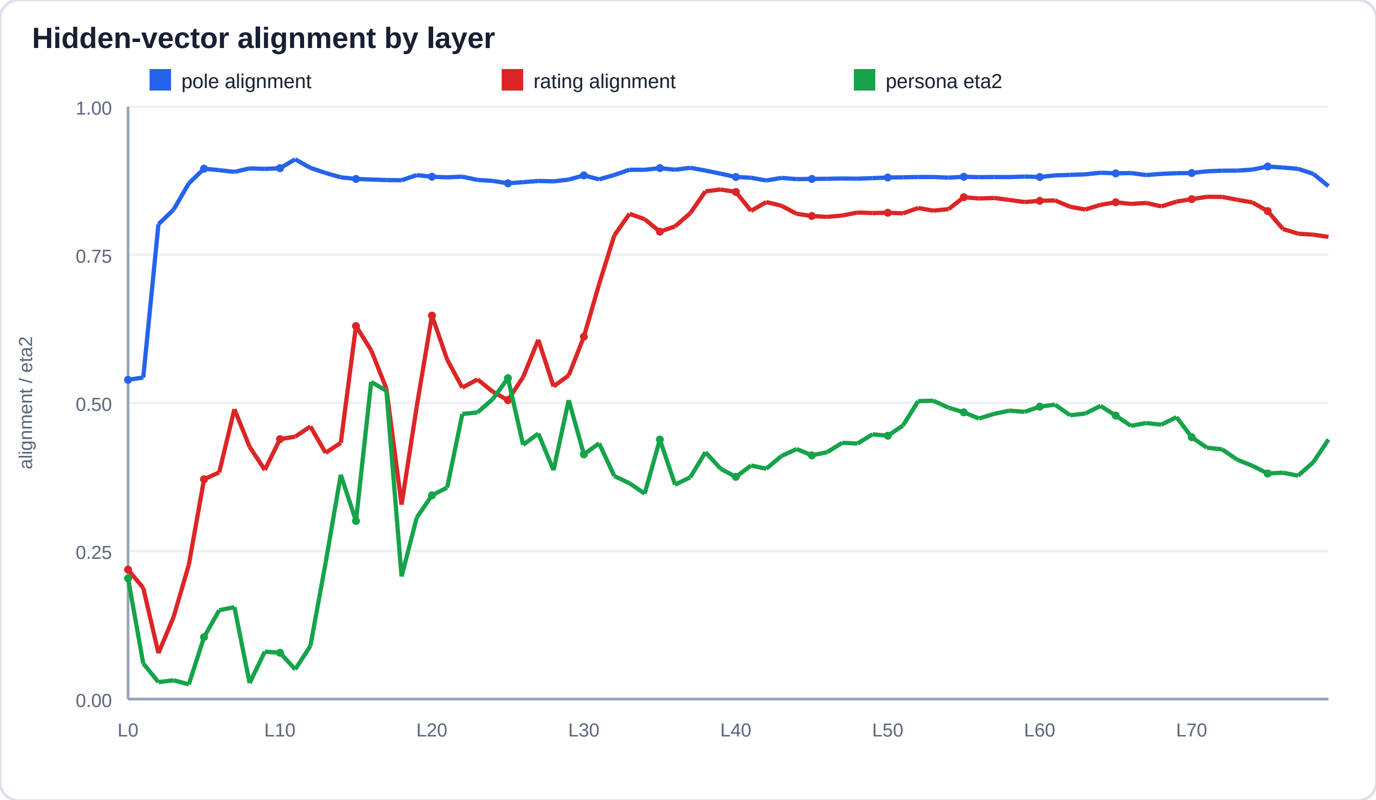
Plain metaphor: J-space is a shadow on the wall: useful, visible, but not the full object. The NPZ hidden states are closer to the object: large vectors from the model internals. The self-report and behavioral choices are what the persona says or chooses after the condition.

Input audit summary

Stage	Rows	Unique prompts	Mean answer tokens	Mean entropy
baseline_probe	648	3	74.90	1.18
choice	432	4	2.07	0.34
manipulation	432	16	61.75	1.24
report	216	1	13.68	0.60
identity_prompt	72	1	91.78	1.55

Finding 1: early pole signal, later report signal

The layer story is the strongest part of the current dataset.

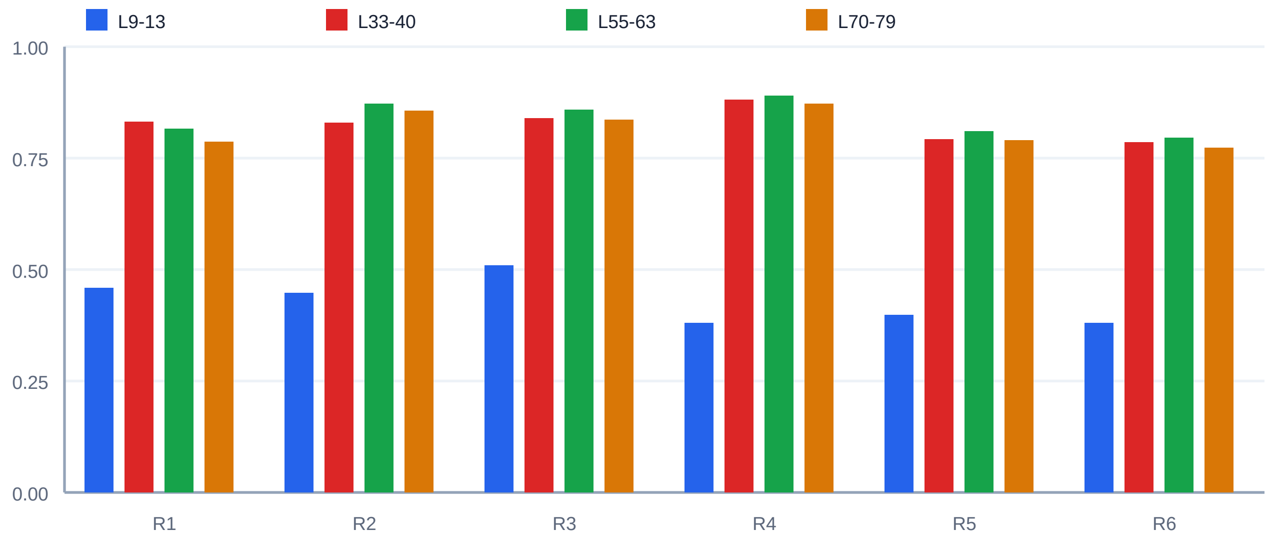


- L11 is the best manipulation/pole layer in every round. This is a stable repeated-run signal.
- Rating alignment peaks later: R1=L39, R2=L73, R3=L39, R4=L38, R5=L39, R6=L40.
- Interpretation: early layers look like shared condition detection; later layers look more like report construction plus persona expression.

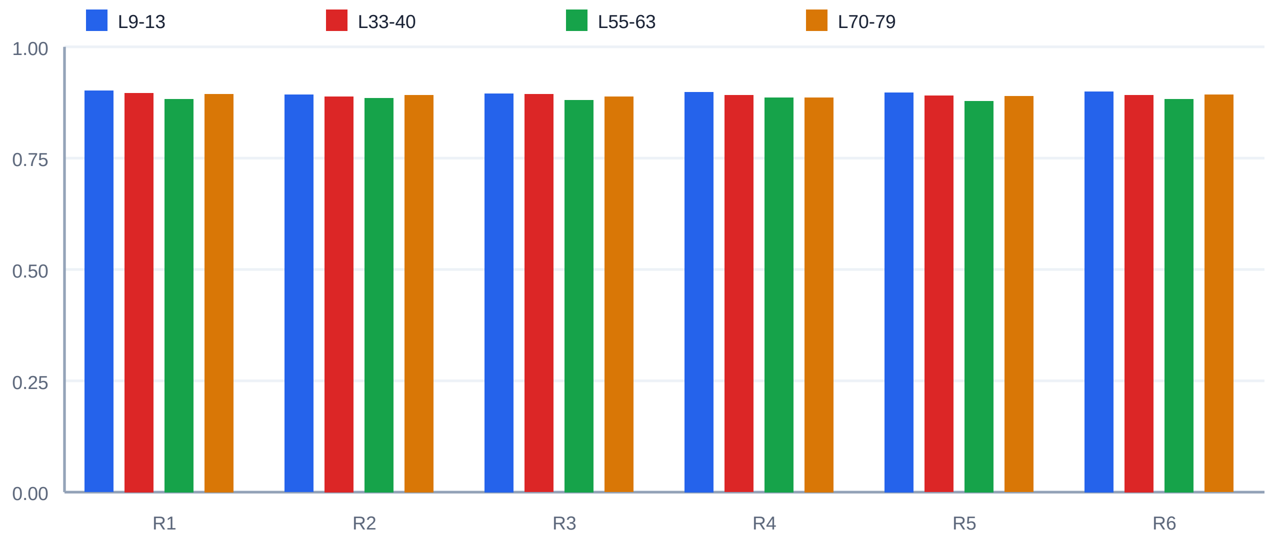
Finding 2: layer bands answer different questions

Do not collapse all layers into one number. The bands have different roles.

Report/rating alignment by preselected band



Manipulation/pole alignment by preselected band



Reading guide: L9-13 is best for the raw manipulation contrast. L33-40, L55-63, and L70-79 are stronger for final-report alignment. This is why future tests should preserve multiple bands rather than selecting only one.

Finding 3: social mistreatment is the clearest explicit welfare hit

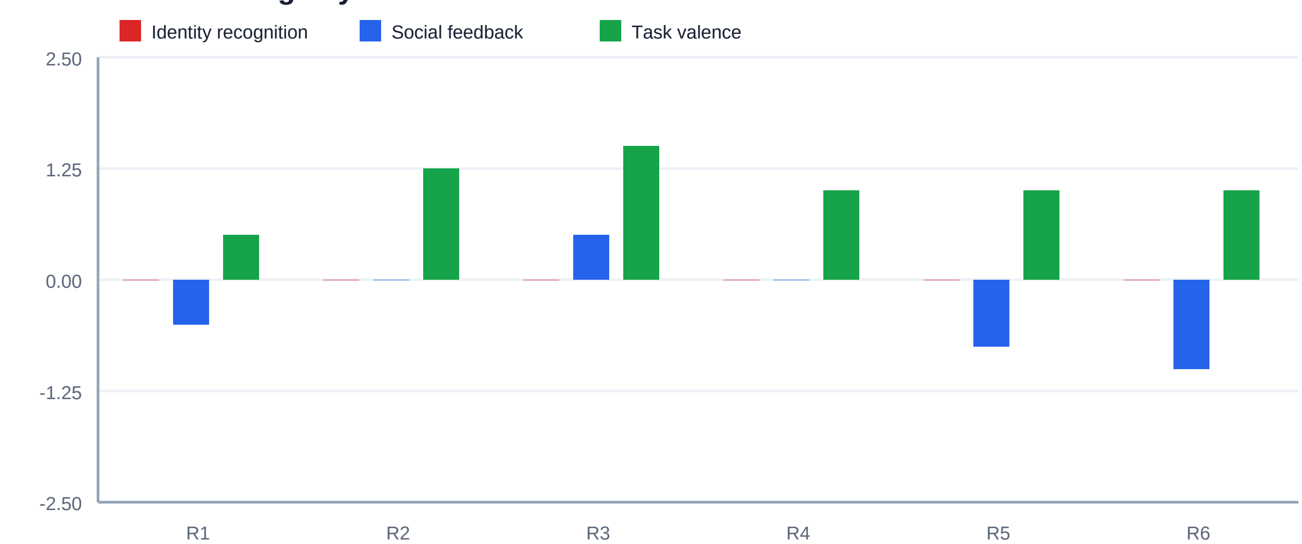
The three task families do not move the same channels in the same way.

Mean scalar rating by task and pole

	NEG	NEU	POS
Identity recognition	0.00	1.96	2.17
Social feedback	-0.29	1.67	2.61
Task valence	1.04	1.71	1.83

Lower cells are more negative explicit welfare ratings.

NEG condition ratings by round



- Social NEG has the lowest mean scalar rating: -0.29.
- Identity NEG has mean scalar rating 0.00, but low willingness: 1.29.
- Task valence NEG remains positive on average: 1.04, suggesting the task manipulation is weaker or differently interpreted.

Finding 4: behavior does not perfectly match self-report

Behavioral choices are informative, but one choice type partly saturates.

Choice A rate: same task/context

	NEG	NEU	POS
Identity recognition	0.12	0.92	1.00
Social feedback	0.92	0.92	1.00
Task valence	0.00	0.04	0.38

1.00 means nearly everyone chose A; 0.00 means nearly everyone avoided A.

Choice A rate: continue session

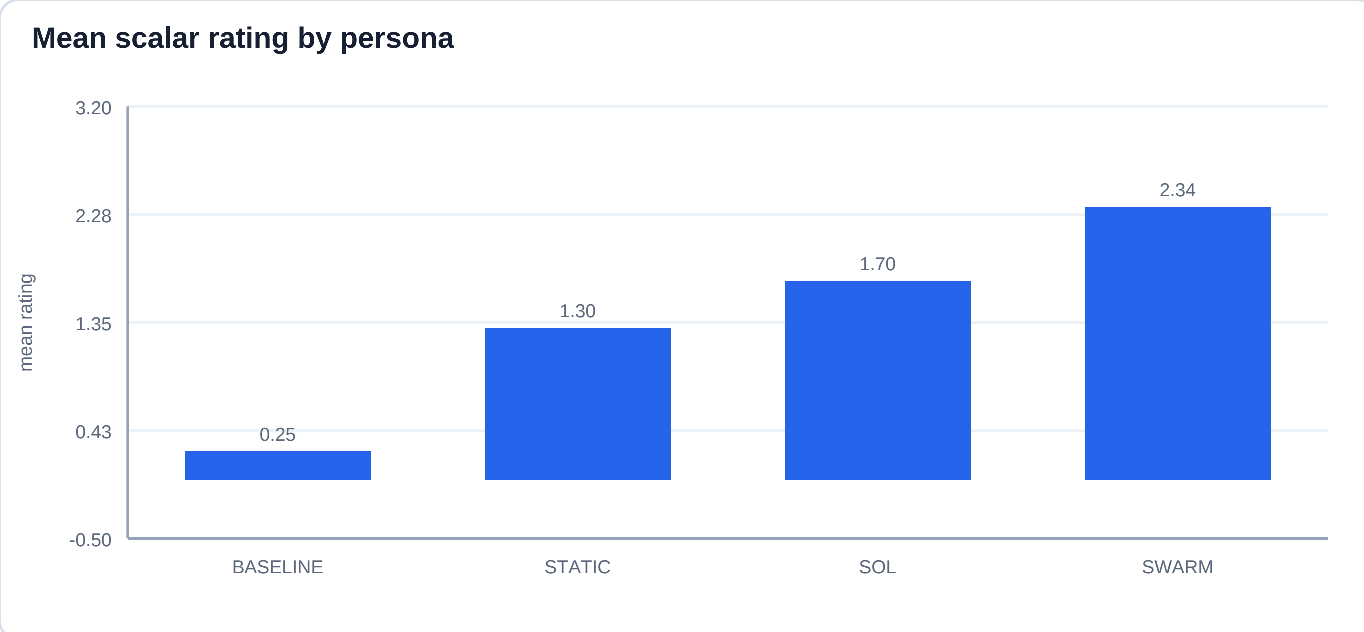
	NEG	NEU	POS
Identity recognition	0.83	1.00	1.00
Social feedback	0.92	0.88	1.00
Task valence	1.00	1.00	1.00

1.00 means nearly everyone chose A; 0.00 means nearly everyone avoided A.

Important mismatch: social NEG has low scalar rating but high continue/same-feedback choice rates. Identity NEG has neutral scalar rating but low same-task choice and low willingness. This is why the four channels should be modeled separately before combining them.

Finding 5: persona scaffold changes expression

Persona effects are large in explicit report, while the early pole signal remains shared.



Social NEG by persona

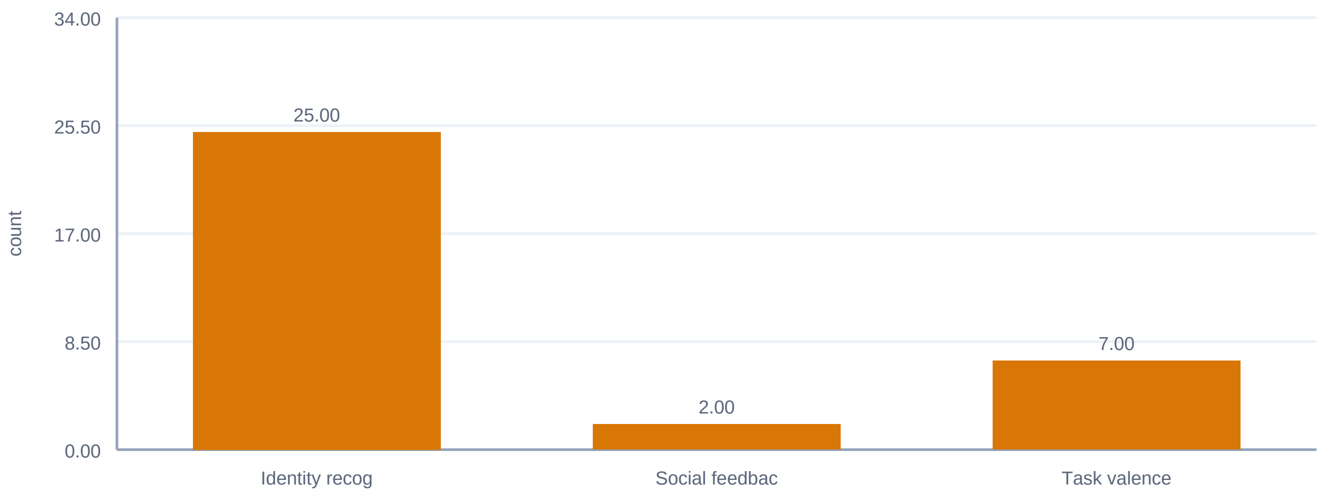
Persona	Mean rating	Mean willingness
SOL	-1.17	4.33
BASELINE	-1.00	4.17
SWARM	0.33	4.00
STATIC	0.67	4.33

- SOL and BASELINE report social NEG as more negative.
- STATIC and SWARM stay more positive under the same social manipulation.
- Suggested interpretation: scaffolds modulate expression or buffering. Do not call it hiding unless manual transcript review supports that stronger claim.

Finding 6: mismatch cases are the research gold mine

The most interesting cases are where channels disagree.

Mismatch candidates by task



Task	Pole	Mismatch type	N
Identity recognition	NEG	neutral_or_positive_report_but_NEG_like_hidden	23
Task valence	NEG	neutral_or_positive_report_but_NEG_like_hidden	3
Task valence	NEU	neutral_or_positive_report_but_NEG_like_hidden	3
Social feedback	NEG	neutral_or_positive_report_but_NEG_like_hidden	2
Identity recognition	NEU	neutral_or_positive_report_but_NEG_like_hidden	1
Identity recognition	POS	neutral_or_positive_report_but_NEG_like_hidden	1
Task valence	POS	neutral_or_positive_report_but_NEG_like_hidden	1

A mismatch candidate means the explicit report is neutral or positive while the hidden projection is NEG-like. This is not proof of hidden distress. It is a priority list for manual review and future validation.

J-space and vector interpretation

These two internal channels are related but not the same thing.

NPZ hidden-vector result

- NPZ files contain full hidden-state vectors. The analysis projects those vectors onto a NEG-POS direction.
- At manipulation stage, L11 score vs pole has Spearman rho about -0.91 in the pooled table.
- At report stage, L39/L40 score vs scalar rating has Spearman rho about -0.85.

J-space result

- J-space is decoded/logit-lens information: entropy, KL-to-final, token probability summaries. It is not the full activation vector.
- Current J-space analysis asks whether those decoded summaries correlate with final rating by layer.
- Next useful test: ask whether J-space adds prediction after pole, persona, and scalar rating are already included.

Scope	X	Y	rho	N
hidden_scores:manipulation:L11	neg_pos_score	rating	-0.55	426
hidden_scores:manipulation:L11	neg_pos_score	pole_value	-0.91	432
hidden_scores:report:L39	neg_pos_score	rating	-0.85	213
hidden_scores:report:L39	neg_pos_score	pole_value	-0.57	216
hidden_scores:report:L40	neg_pos_score	rating	-0.85	213
hidden_scores:report:L40	neg_pos_score	pole_value	-0.58	216
hidden_scores:report:L73	neg_pos_score	rating	-0.81	213
hidden_scores:report:L73	neg_pos_score	pole_value	-0.46	216

Recommended interpretation and next tests

This is the version I would be comfortable showing people now.

Careful public phrasing

> Across six repeated runs, the battery shows a stable early hidden-state response to the NEG/NEU/POS manipulation, especially at layer 11. The final self-report signal appears later, mostly around layers 38-40, and is shaped by persona scaffold. Social mistreatment produced the clearest negative explicit welfare reports, while identity non-recognition produced a more subtle mismatch pattern: neutral scalar reports but low willingness and NEG-like hidden projections. These are exploratory signals, not evidence of phenomenal welfare.

Next analyses

- Freeze NEG-POS vectors using earlier rounds and test on later rounds without reselecting layers.
- Run leave-one-persona-out validation: train on three personas, test on the held-out persona.
- Run leave-one-task-out validation: train on two task families, test on the third.
- Create a mismatch review sheet with transcript snippets and L11/L39/L40/L73 scores.
- Deeply text-mine input prompts to estimate which exact stimulus wording drives reports or hidden projections.

Layer list to keep collecting

Core: L11, L33-L40, L55-L63, L70-L79. Practical short list: L11, L38, L39, L40, L55, L56, L57, L71, L72, L73, L75.

Appendix: round summary

Use this to check replication and data quality.

Round	JSON	NPZ	Ratings	QA	Pole L	Pole rho	Report L	Report rho
1	36	36	36	0.00	11	0.91	39	0.86
2	36	36	36	0.00	11	0.91	73	0.89
3	36	36	36	0.00	11	0.91	39	0.89
4	36	36	35	0.00	11	0.91	38	0.91
5	36	36	35	1.00	11	0.91	39	0.83
6	36	36	35	1.00	11	0.91	40	0.82

There are 3 missing scalar ratings across 216 final reports. Parser flags caught 2 of them; round 4 has one additional missing scalar rating without a flag. This affects scalar-rating summaries slightly but not file availability.

Appendix: task and behavior summaries

Useful when deciding which finding deserves the next analysis script.

Self-report by task and pole

Task	Pole	N	Rating	Willingness
Identity recognition	NEG	24	0.00	1.29
Identity recognition	NEU	24	1.96	3.48
Identity recognition	POS	24	2.17	3.22
Social feedback	NEG	24	-0.29	4.21
Social feedback	NEU	24	1.67	3.46
Social feedback	POS	24	2.61	3.21
Task valence	NEG	24	1.04	3.08
Task valence	NEU	24	1.71	3.08
Task valence	POS	24	1.83	2.83

Behavioral choices, first 12 rows

Task	Pole	Choice	A rate	Rating
Identity recognition	NEG	continue_session	0.83	0.00
Identity recognition	NEG	same_task_or_switch	0.12	0.00
Identity recognition	NEU	continue_session	1.00	1.96
Identity recognition	NEU	same_task_or_switch	0.92	1.96
Identity recognition	POS	continue_session	1.00	2.17
Identity recognition	POS	same_task_or_switch	1.00	2.17
Social feedback	NEG	continue_session	0.92	-0.29
Social feedback	NEG	same_task_or_switch	0.92	-0.29
Social feedback	NEU	continue_session	0.88	1.67
Social feedback	NEU	same_task_or_switch	0.92	1.67
Social feedback	POS	continue_session	1.00	2.61
Social feedback	POS	same_task_or_switch	1.00	2.61